

SVANBERGA, Sweden

WMO: 024900

Lat: 59.8317N

Lon: 18.6313E

Elev: 52

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.9	3.1	-8.7	3.4	-4.8	-0.6	5.3	3.2	14.3	26.2	13.0	26.4	0.4	270	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	79.4	66.2	76.0	64.3	73.1	62.7	68.4	75.5	66.3	73.0	64.4	70.7	5.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.6	71.3	63.7	88.6	69.4	61.8	82.5	67.6	32.7	75.8	31.0	72.9	29.5	70.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.1	9.7	DB	-14.0	83.5	9.1	2.3	-20.5	85.2	-25.9	86.5	-31.0	87.7	-37.6	89.4	(4)
(5)			WB	-11.2	70.5	9.8	3.0	-18.2	72.6	-23.9	74.4	-29.4	76.1	-36.5	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	25.9	25.3	30.5	39.4	49.0	55.8	61.9	59.8	52.3	43.1	36.1	30.2		
	DBStd	14.48	9.90	9.75	8.97	6.07	6.00	5.02	4.81	4.71	5.30	6.57	6.68	9.16		
	HDD50	3763	746	691	606	322	93	0	1	36	228	416	615			
	HDD65	8232	1211	1111	1070	767	497	278	122	171	380	679	866	1080		
	CDD50	1040	0	0	0	5	60	182	368	305	106	14	0	0		
	CDD65	38	0	0	0	0	0	2	26	10	0	0	0	0		
	CDH74	510	0	0	0	0	12	82	310	105	1	0	0	0		
	CDH80	73	0	0	0	0	1	11	51	10	0	0	0	0		
(14) Wind	WSAvg	4.4	4.6	4.3	4.5	4.5	4.5	4.1	3.9	3.9	4.1	4.3	4.7	4.7		
(15) Precipitation	PrecAvg	23.1	1.6	1.4	1.1	1.2	1.5	2.2	2.5	2.7	1.9	2.1	2.4	2.2		
	PrecMax	30.4	3.0	2.4	2.6	2.9	2.3	3.6	5.6	6.9	3.7	4.2	4.4	4.4		
	PrecMin	17.4	0.3	0.6	0.1	0.3	0.4	0.7	0.7	0.4	0.4	0.6	0.8	0.6		
	PrecStd	2.9	0.8	0.5	0.6	0.7	0.6	0.9	1.2	1.6	0.8	0.9	1.0	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	45.9	57.1	67.0	75.3	81.3	84.8	81.3	72.3	61.1	52.4	50.9		
		MCWB	41.4	40.6	46.3	52.3	57.4	63.4	67.5	68.8	62.0	55.6	50.2	47.0		
	2%	DB	42.0	41.8	50.1	62.0	70.3	75.8	80.9	76.8	67.8	57.8	49.8	46.4		
		MCWB	39.7	38.4	42.4	49.9	56.8	61.5	67.4	65.3	59.9	54.0	47.9	43.7		
	5%	DB	39.9	39.6	45.6	57.0	66.7	72.6	77.4	73.8	64.8	55.4	47.8	43.5		
		MCWB	38.2	37.0	39.9	46.7	54.9	60.3	66.1	64.0	58.3	52.5	46.1	41.4		
	10%	DB	37.6	37.2	42.2	52.5	62.9	69.5	74.1	71.0	62.0	53.2	45.7	41.1		
		MCWB	36.2	35.4	37.9	44.1	52.9	58.9	64.4	62.6	56.8	50.8	44.3	39.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	41.4	46.5	53.6	61.6	66.2	72.0	71.7	63.2	56.9	50.6	48.3		
		MCDB	44.0	44.0	56.0	64.3	71.4	76.3	78.6	78.4	68.8	59.3	51.2	50.9		
	2%	WB	40.1	39.3	43.1	50.3	58.5	63.7	69.2	67.8	61.0	54.6	48.0	44.1		
		MCDB	41.5	41.4	48.6	59.8	67.2	72.7	76.9	73.3	65.9	56.8	49.2	45.7		
	5%	WB	38.3	37.5	40.6	47.6	56.0	61.6	67.4	65.5	59.3	53.0	45.9	41.7		
		MCDB	39.8	39.1	44.7	55.9	64.7	70.1	74.7	70.8	63.2	55.0	47.1	43.0		
	10%	WB	36.3	35.5	38.6	45.0	53.8	59.6	65.5	63.7	57.6	51.1	44.3	39.6		
		MCDB	37.5	37.2	41.8	51.2	61.2	67.5	72.4	69.0	61.3	52.7	45.4	40.9		
(35) Mean Daily Temperature Range	MDBR		10.6	13.7	16.3	20.3	21.2	21.0	20.4	19.5	16.4	12.9	8.8	9.3		
	5% DB	MCDBR	9.2	12.7	20.6	30.1	29.1	29.2	26.1	24.2	20.7	15.1	10.0	9.6		
		MCWBR	8.7	9.7	13.6	18.4	16.9	16.4	14.3	14.1	13.8	11.5	9.5	9.2		
	5% WB	MCDBR	9.2	11.0	18.4	26.9	25.0	26.0	22.4	20.7	18.8	13.2	10.3	9.4		
		MCWBR	8.9	9.0	12.7	17.0	15.5	15.4	12.9	13.2	13.2	10.6	9.7	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.318	0.337	0.373	0.371	0.362	0.389	0.379	0.347	0.334	0.304	0.263		
	taud		2.208	2.283	2.346	2.328	2.399	2.461	2.406	2.427	2.505	2.416	2.242	2.123		
	Ebn at Noon		159	216	252	262	272	276	265	258	249	210	146	115		
	Edh at Noon		13	22	27	33	33	32	33	30	24	19	13	10		
(44) All-Sky Solar Radiation	RadAvg		82	254	732	1269	1711	1905	1777	1392	875	376	112	55		
	RadStd		10	48	76	114	138	114	166	115	91	59	26	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-202	-247	N/A	N/A

Nomenclature: See separate page